Recurrent Models for Rumor Detection on Social Media

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Abstract

Rumors spread through social media in sequences of posts. We model these sequences with LSTM and GRU networks to detect rumors early. The best model reaches 92.8 percent accuracy.

Keywords: rumor detection, social media, recurrent neural network, misinformation

1. Introduction

Rumor detection on social media requires modelling how posts unfold over time. We study recurrent neural networks for this task.

2. Related Work

Convolutional approaches to fake news such as Convolutional Neural Networks for Fake News Classification ignore temporal structure.

3. Methodology

Each rumor is represented as a time-ordered sequence of posts. We train LSTM and GRU networks and compare them with a feed-forward baseline.

4. Results

The LSTM reached 92.8 percent accuracy and the GRU 91.9 percent. Both beat the baseline by about four points.

5. Limitations

The models struggle with multilingual data. Training has a high computational cost compared with CNNs.

6. Conclusion

Recurrent models capture temporal rumor patterns, but cost and language coverage remain obstacles.

7. Future Work

Future work should add attention mechanisms and test real-time deployment.

References

[1] Convolutional Neural Networks for Fake News Classification, Journal of Demonstration Studies, 2019.

Note: this is a SYNTHETIC demonstration paper with fictional authors and invented results, created to demo ResearchGraph. It is not a real publication.